Residual Interference Analysis of Nulling for Terrestrial–Satellite Downlink Coexistence: Channel Estimation Errors and SNR Asymptotics

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***Abstract*—The upper mid-band FR3 spectrum (7–24 GHz) has garnered significant interest for future cellular services. However, utilizing a large portion of this band requires careful interference coordination with incumbent satellite systems. This paper investigates interference from high-power terrestrial base stations (TN-BSs) to satellite downlink receivers. A central challenge is that the victim receivers, i.e., ground-based non-terrestrial user equipment (NTN-UEs), such as satellite customer premises equipment, must first be detected, and their channels estimated, before the TN-BS can effectively place nulls in their directions. We explore a potential solution where NTN-UEs periodically transmit preambles or beacon signals that TN-BSs can use for detection and channel estimatio. The performance of this nulling approach is analyzed in a simplified scenario with a single victim, revealing the interplay between path loss and estimation quality in determining nulling performance. To further validate the method, we conduct a detailed multi-user site-specific ray-tracing (RT) simulation in a rural environment. The results show that the proposed nulling approach is effective under realistic parameters, even with high densities of victim units, although TN-BS may require a substantial number of antennas.**

I. Introduction

The FR3 upper mid-band spectrum (7–24 GHz) offers unique value for cellular networks by balancing coverage and capacity needs in 5G and 6G technologies [1], [2], and has been identified by the 2023 World Radiocommunication Conference (WRC-23) as an essential band for future expansion [3]. A key challenge arises as FR3 overlaps with the satellite Ku-band (12–18 GHz), where commercial satellite systems operate as incumbents. The Federal Communications Commission (FCC) has proposed spectrum sharing rules to enable coexistence between terrestrial networks (TN) and non-terrestrial networks (NTN) via interference mitigation protocols [4]. Recent theoretical work further shows such coexistence is feasible under clustered TN deployments using tractable outage models [5].

Previous research has extensively addressed interference mitigation, particularly focusing on the terrestrial-to-satellite uplink scenario [6]. Explored techniques include interference nulling from TN base stations (TN-BSs) to satellites [6], 3D beamforming with feed array reflectors for spectral coexistence [7], using transmit beamforming at the TN-BS to protect satellite terminals by leveraging their fixed pointing directions [8], and secure beamforming for 5G mmWave systems coexisting with satellite networks [9]. Other efforts introduced theoretical interference models [10] and multicast beamforming strategies for cloud-based terrestrial-satellite architectures [11]. More recently, interference management in coexisting non-terrestrial networks (NTNs), such as dense low Earth orbit (LEO) constellations, has been studied. These include in-band coexistence via strategic satellite selection by secondary LEO systems to protect incumbents [12], optimization and learning-based selection frameworks to maximize secondary system capacity under interference constraints [13].While these studies address uplink and inter-satellite interference, recent work has also examined the downlink case, notably proposing reverse spectrum pairing [14], [15], in which TN uplink overlaps with NTN downlink on time. This paper focuses on the distinct challenge of satellite downlink interference, where TN-BS transmissions interfere with NTN-UEs.

Satellite downlink presents fundamentally different challenges compared to uplink interference scenarios. Potential victim NTN-UEs remain unknown to TN-BS operators, with locations typically falling within the TN-BS antenna downtilt regions. NTN-UE channel information remains equally unknown, significantly complicating interference management. In contrast, satellite uplink interference management benefits from deterministic satellite positions available through ephemeris data. Satellites maintain relatively constant line-of-sight (LOS) paths, enabling easier detection and interference mitigation compared to the dynamic, unknown NTN-UE distribution in downlink scenarios. Interference mitigation in the satellite downlink thus requires detecting potential victims and estimating the channels to those victims before interference nulling.

To address this challenge, our work introduces a joint beamforming optimization framework for TN-to-NTN interference mitigation, focusing specifically on TN-BS interference to NTN-UEs. The framework estimates interference channels to multiple NTN-UEs without precise location knowledge, enabling spatial nulling while preserving TN signal quality.

Key contributions include:

- *Detection and Nulling protocol:* We consider a four-stage protocol shown in Fig. 1 where the potential NTN-UEs

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broadcast beacon or preamble signals. From the preamble signals, the TN-gNB can detect the victim NTN-UEs and estimate the downlink channel. After detecting and estimating the channels to the victims, the TN-gNB can beamform to the desired users with nulls in the directions to the victims.

- *Path loss effect on channel estimation:* Perfect channel knowledge enables complete signal nulling to victims. For imperfect estimation, path loss creates competing effects: increasing path loss improves victim detectability and channel estimation quality, while simultaneously amplifying residual interference impact. Analysis reveals critical detection-interference tradeoffs in single-user scenarios.
- *Case Study:* We validate the performance in a rural area near Boulder, Colorado with realistic ray tracing simulations capturing complex propagation phenomena in rural macrocell (RMa) environments. Results confirm interference reduction effectiveness across varying antenna configurations and victim positions.
- *Massive MIMO necessity:* Analysis demonstrates minimum antenna requirements (64+ elements) for effective null steering toward NTN-UEs while maintaining adequate TN-UE signal quality. Increasing antenna counts significantly enhances coexistence performance.

II. System Model

A. Network Model

A terrestrial base station, denoted by TX, sends a downlink signal to an intended terrestrial user, RX_0. The same transmission may also reach K other receivers, $\mathrm{RX}_1, \ldots, \mathrm{RX}_K$, for which it is unwanted interference. We call these receivers *victims*. They may be user terminals or base stations belonging to a non-terrestrial network, located on the ground or in the air.

TX has N_{tx} transmit antennas and each receiver has one antenna. The channel vector $\boldsymbol{h}_i \in \mathbb{C}^{N_{\mathrm{tx}}}$ describes the link from TX to RX_i: its entries encode the amplitude and phase responses associated with the transmit antennas. The beamforming vector $\boldsymbol{w}$ specifies how the transmitted signal is weighted across these antennas. We impose $\|\boldsymbol{w}\| = 1$ so that changing the beam changes its spatial distribution while keeping the transmitted energy fixed.

Let $\mathcal{E}_{\mathrm{D}}$ be the downlink energy per symbol and N_0 the noise energy at each receiver, assumed common to all receivers. The squared inner product $|\boldsymbol{w}^H \boldsymbol{h}_i|^2$ determines how strongly the beam couples to receiver i. At the intended receiver this is useful signal, giving the signal-to-noise ratio

$$\mathsf{SNR}_0 = \frac{\mathcal{E}_{\mathrm{D}}}{N_0} |\boldsymbol{w}^H \boldsymbol{h}_0|^2. \tag{1}$$

At a victim receiver the same transmitted signal is interference, giving the interference-to-noise ratio

$$\mathsf{INR}_i = \frac{\mathcal{E}_{\mathrm{D}}}{N_0} |\boldsymbol{w}^H \boldsymbol{h}_i|^2, \qquad i = 1, \ldots, K. \tag{2}$$

The beam should preserve useful signal at RX_0 while reducing these unwanted couplings.

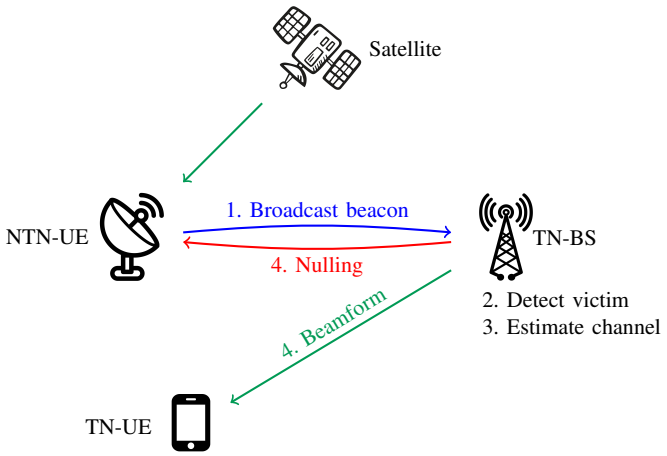


Fig. 1. Proposed interference mitigation framework for non-terrestrial and terrestrial co-existence networks: (1) NTN-UE transmits beacon signal, (2) TN-BS detects potential interference victim, (3) TN-BS estimates channel between TN-BS and NTN-UE, and (4) TN-BS implements spatial filtering with simultaneous null steering toward NTN-UE and optimized beamforming to TN-UE, maximizing spectrum sharing efficiency.

B. Uplink Training and Channel Estimation

To estimate a victim's channel, TX listens to a known training sequence sent in the reverse direction. We assume channel reciprocity: the channel measured in this uplink is also the channel relevant to downlink interference. For known sequence timing, correlating the received samples with the training sequence produces the observation

$$\begin{cases} \boldsymbol{r}_i = a_i \boldsymbol{h}_i + \boldsymbol{z}_i, \\ \boldsymbol{z}_i \sim \mathcal{CN}(\boldsymbol{0}, N_0 \boldsymbol{I}). \end{cases} \tag{3}$$

Here $a_i \neq 0$ is the known training coefficient and $\boldsymbol{z}_i$ is zero-mean complex Gaussian noise with independent antenna components. The total training energy is $\mathcal{E}_{\mathrm{U}} = |a_i|^2$. It may be much larger than $\mathcal{E}_{\mathrm{D}}$ because the training sequence contains many samples, while $\mathcal{E}_{\mathrm{D}}$ refers to one downlink symbol.

The average channel gain per transmit antenna is

$$G_i = \frac{\|\boldsymbol{h}_i\|^2}{N_{\mathrm{tx}}}. \tag{4}$$

The corresponding training SNR per antenna is

$$\gamma_i^{\mathrm{U}} = \frac{\mathcal{E}_{\mathrm{U}} G_i}{N_0}. \tag{5}$$

Dividing the observation by the known coefficient removes the training signal's amplitude and phase:

$$\widehat{\boldsymbol{h}}_i = \frac{a_i^*}{|a_i|^2} \boldsymbol{r}_i = \boldsymbol{h}_i + \boldsymbol{v}_i. \tag{6}$$

The estimation error is scaled observation noise, $\boldsymbol{v}_i = (a_i^*/|a_i|^2)\boldsymbol{z}_i$, and therefore

$$\boldsymbol{v}_i \sim \mathcal{CN}\left(\boldsymbol{0}, \frac{N_0}{\mathcal{E}_{\mathrm{U}}} \boldsymbol{I}\right). \tag{7}$$

This relation explicitly connects the amount of training energy to the uncertainty in the channel estimate.

C. Estimated-Channel Nulling

Assume perfect knowledge of the intended channel $\boldsymbol{h}_0$, and define its unit direction by

$$\tilde{\boldsymbol{h}}_0 = \frac{\boldsymbol{h}_0}{\|\boldsymbol{h}_0\|}. \tag{8}$$

Using the victim-channel estimates, choose the beamformer to maximize useful coupling while penalizing estimated interference:

$$\widehat{\boldsymbol{w}} = \arg\max_{\|\boldsymbol{w}\|=1} \left[|\boldsymbol{w}^H \tilde{\boldsymbol{h}}_0|^2 - \lambda \sum_{i=1}^{K} |\boldsymbol{w}^H \widehat{\boldsymbol{h}}_i|^2 \right]. \tag{9}$$

The parameter $\lambda \geq 0$ sets the relative weight of interference suppression. For finite λ, the objective is the quadratic form $\boldsymbol{w}^H \boldsymbol{Q} \boldsymbol{w}$, where

$$\boldsymbol{Q} = \tilde{\boldsymbol{h}}_0 \tilde{\boldsymbol{h}}_0^H - \lambda \sum_{i=1}^{K} \widehat{\boldsymbol{h}}_i \widehat{\boldsymbol{h}}_i^H. \tag{10}$$

Thus an eigenvector associated with the largest eigenvalue of $\boldsymbol{Q}$ solves the optimization.

The next section specializes to one victim and the limit $\lambda \to \infty$. In this case, called *hard nulling*, we impose $\boldsymbol{w}^H \widehat{\boldsymbol{h}}_1 = 0$ exactly. This removes the coupling to the estimated channel; because the estimate contains error, some coupling to the true channel may remain.

III. Residual Interference with Channel Estimation Error

We derive the single-victim residual INR from the channel-estimation model in (6)–(7). The derivation first preserves the exact cancellation in the leakage amplitude, then evaluates its numerator moments, and finally states the approximations used to obtain a closed-form mean-INR estimate. The same derivation explains the linear growth at weak channel gain and the plateau at strong channel gain, including their conditions and finite-array limits. A small open square below an equation number marks a key result.

A. Single-Victim Model and Exact Projection Identities

Consider one intended receiver and one victim receiver, with fixed channel vectors $\boldsymbol{h}_0$ and $\boldsymbol{h}_1$. Assume $N_{\mathrm{tx}} \geq 2$ and perfect knowledge of $\boldsymbol{h}_0$. The transmitter chooses a unit vector $\widehat{\boldsymbol{w}}$ that is orthogonal to the *estimated* victim channel. Our objective is to compute the average power that still reaches the victim through the true channel, namely $\mathbb{E}[\mathsf{INR}_1]$. All expectations below are over the channel-estimation error, with both true channels held fixed.

Equivalently, the beamformer solves

$$\begin{aligned} &\max_{\boldsymbol{w}} \ |\boldsymbol{w}^H \boldsymbol{h}_0|^2 \\ &\text{subject to} \quad \boldsymbol{w}^H \widehat{\boldsymbol{h}}_1 = 0, \quad \|\boldsymbol{w}\| = 1. \end{aligned} \tag{11}$$

Estimation error and channel strength: The estimate differs from the true victim channel by additive noise:

$$\begin{cases} \widehat{\boldsymbol{h}}_1 = \boldsymbol{h}_1 + \boldsymbol{v}_1, \\ \boldsymbol{v}_1 \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}), \\ \sigma^2 = \dfrac{N_0}{\mathcal{E}_{\mathsf{U}}}. \end{cases} \tag{12}$$

Here σ^2 is the error variance per antenna; increasing the training energy $\mathcal{E}_{\mathsf{U}}$ reduces this variance.

The scalar

$$H = \|\boldsymbol{h}_1\|^2 = N_{\mathrm{tx}} G_1 \tag{13}$$

measures the total squared channel magnitude across the transmit antennas. We initially assume $H > 0$ and $\boldsymbol{h}_0 \neq \boldsymbol{0}$; zero victim-channel gain is recovered by continuity. The training SNR per antenna can now be written as

$$\gamma_1^{\mathsf{U}} = \frac{G_1}{\sigma^2} = \frac{H}{N_{\mathrm{tx}} \sigma^2}. \tag{14}$$

We also define the energy ratio $k = \mathcal{E}_{\mathsf{D}} / \mathcal{E}_{\mathsf{U}}$, which sets the scale of the residual INR.

Relative channel directions: The normalized inner product measures how closely the desired and victim channels are aligned:

$$\begin{cases} \rho = \dfrac{\boldsymbol{h}_1^H \boldsymbol{h}_0}{\|\boldsymbol{h}_1\| \, \|\boldsymbol{h}_0\|}, \\ r = |\rho|^2 \in [0, 1]. \end{cases} \tag{15}$$

A common phase rotation of a beamforming vector does not change any received power. We use this freedom to define

$$\boldsymbol{u} = e^{-j\phi} \frac{\boldsymbol{h}_0}{\|\boldsymbol{h}_0\|}, \tag{16}$$

where $\phi = \arg(\boldsymbol{h}_1^H \boldsymbol{h}_0)$ when the inner product is nonzero, and $\phi = 0$ otherwise. Then $\|\boldsymbol{u}\| = 1$ and $\boldsymbol{u}^H \boldsymbol{h}_1 = \sqrt{Hr}$ is real and nonnegative. This phase convention will simplify the coordinate decomposition.

Projection and normalization: The matrix

$$\boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} = \boldsymbol{I} - \frac{\widehat{\boldsymbol{h}}_1 \widehat{\boldsymbol{h}}_1^H}{\|\widehat{\boldsymbol{h}}_1\|^2} \tag{17}$$

removes the component of any vector along $\widehat{\boldsymbol{h}}_1$. Applying it to $\boldsymbol{u}$ keeps the part of the desired direction that satisfies the nulling constraint. Its squared length is

$$\begin{aligned} p &= \|\boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{u}\|^2 \\ &= 1 - \frac{|\boldsymbol{u}^H \widehat{\boldsymbol{h}}_1|^2}{\|\widehat{\boldsymbol{h}}_1\|^2}. \end{aligned} \tag{18}$$

Because the projector is Hermitian and idempotent, $p = \boldsymbol{u}^H \boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{u}$ as well. Normalizing the projected vector gives the beamformer

$$\widehat{\boldsymbol{w}} = \frac{\boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{u}}{\sqrt{p}}. \tag{19}$$

It maximizes the desired-channel power among unit vectors satisfying the nulling constraint: for any feasible $\boldsymbol{w}$, $|\boldsymbol{w}^H \boldsymbol{u}| = |\boldsymbol{w}^H \boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{u}| \leq \sqrt{p}$, with equality for (19). For $N_{\mathrm{tx}} \geq 2$ and $\sigma^2 > 0$, the exceptional event $p = 0$ has probability zero.

Residual interference: Define the leakage amplitude before beam normalization by

$$F = \boldsymbol{u}^H \boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{h}_1. \tag{20}$$

The actual amplitude at the victim is $\widehat{\boldsymbol{w}}^H \boldsymbol{h}_1 = F / \sqrt{p}$. The mean INR is therefore

$$\mathbb{E}[\mathsf{INR}_1] = \frac{\mathcal{E}_{\mathsf{D}}}{N_0} \mathbb{E}\left[\frac{|F|^2}{p} \right]. \tag{21}$$

The projector annihilates $\widehat{\boldsymbol{h}}_1 = \boldsymbol{h}_1 + \boldsymbol{v}_1$, which gives the exact identities

$$\begin{cases} \boldsymbol{P}_{\widehat{h}_1}^{\perp} \boldsymbol{h}_1 = -\boldsymbol{P}_{\widehat{h}_1}^{\perp} \boldsymbol{v}_1, \\ \widehat{\boldsymbol{w}}^H \boldsymbol{h}_1 = -\widehat{\boldsymbol{w}}^H \boldsymbol{v}_1. \end{cases} \tag{22}$$

Thus the residual interference is determined by the estimation error projected onto the selected beam. The beam itself depends on that error, so these two random quantities cannot be treated as independent. Nevertheless, the unit norm of $\widehat{\boldsymbol{w}}$ gives

$$\begin{cases} |\widehat{\boldsymbol{w}}^H \boldsymbol{h}_1|^2 \le \|\boldsymbol{v}_1\|^2, \\ 0 \le \mathbb{E}[\mathsf{INR}_1] \le k N_{\mathrm{tx}}. \end{cases} \tag{23}$$

The second line follows from $\mathbb{E}\|\boldsymbol{v}_1\|^2 = N_{\mathrm{tx}}\sigma^2$ and the definitions of σ^2 and k.

B. Exact Cancellation Before Taking Moments

Choose orthonormal directions $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ so that

$$\begin{cases} \boldsymbol{h}_1 = \sqrt{H}\, \boldsymbol{e}_1, \\ \boldsymbol{u} = \sqrt{r}\, \boldsymbol{e}_1 + \sqrt{1-r}\, \boldsymbol{e}_2. \end{cases} \tag{24}$$

For $r = 1$, $\boldsymbol{e}_2$ can be any unit vector orthogonal to $\boldsymbol{e}_1$. Write the estimation error as $\boldsymbol{v}_1 = \zeta \boldsymbol{e}_1 + \boldsymbol{\xi}$, where $\zeta = \boldsymbol{e}_1^H \boldsymbol{v}_1$ and $\boldsymbol{e}_1^H \boldsymbol{\xi} = 0$. Define $x = \boldsymbol{e}_2^H \boldsymbol{\xi}$, $Q = \|\boldsymbol{\xi}\|^2$, and $c = \sqrt{H} + \zeta$, giving $\widehat{\boldsymbol{h}}_1 = c\boldsymbol{e}_1 + \boldsymbol{\xi}$. The variables ζ and x are independent $\mathcal{CN}(0, \sigma^2)$ scalars. Also, c is independent of (x, Q), whereas Q includes $|x|^2$ and is therefore dependent on x.

The estimated-channel energy is

$$T_{\mathrm{h}} = \|\widehat{\boldsymbol{h}}_1\|^2 = |c|^2 + Q. \tag{25}$$

Its energy after projection orthogonal to $\boldsymbol{u}$ is

$$U = T_{\mathrm{h}} - |\boldsymbol{u}^H \widehat{\boldsymbol{h}}_1|^2. \tag{26}$$

Thus the beam-normalization factor is $p = U/T_{\mathrm{h}}$. Substituting these coordinates into the leakage amplitude gives

$$\begin{aligned} F &= \sqrt{Hr} - \frac{(\sqrt{r}\, c + \sqrt{1-r}\, x) c^* \sqrt{H}}{T_{\mathrm{h}}} \\ &= \frac{\sqrt{H}}{T_{\mathrm{h}}} \left[\sqrt{r}\, Q - \sqrt{1-r}\, x c^* \right]. \end{aligned} \tag{27}$$

The terms proportional to $\sqrt{r}\,|c|^2$ cancel exactly for every error realization. Retaining this cancellation before taking moments preserves the dependence between the channel and error contributions. Squaring and dividing by p yields the exact residual power

$$|\widehat{\boldsymbol{w}}^H \boldsymbol{h}_1|^2 = \frac{H |\sqrt{r}\, Q - \sqrt{1-r}\, x c^*|^2}{T_{\mathrm{h}} U}. \tag{28}$$

This expression holds for every $N_{\mathrm{tx}} \ge 2$, without any moment approximation. Appendix A provides the coordinate construction, detailed cancellation, and an equivalent scalar representation for Monte Carlo averaging.

C. Mean Residual INR

Let the numerator of (28) be

$$\mathcal{N} \triangleq H \left| \sqrt{r}\, Q - \sqrt{1-r}\, x c^* \right|^2. \tag{29}$$

The exact moments of $\mathcal{N}$, T_{h}, and U are evaluated in Appendix A. Two subsequent approximations give a closed-form estimate of the mean INR.

First approximation: Replace each random denominator energy by its own mean, while retaining the exact numerator mean:

$$\mathbb{E}\left[\frac{\mathcal{N}}{T_{\mathrm{h}} U} \right] \approx \frac{\mathbb{E}\mathcal{N}}{(\mathbb{E}T_{\mathrm{h}})(\mathbb{E}U)}. \tag{30}$$

At fixed $\gamma_1^{\mathsf{U}} > 0$ and r, the relative variances of both denominator energies are $O(N_{\mathrm{tx}}^{-1})$; their explicit expressions are given in Appendix A. This motivates the replacement after the exact amplitude cancellation. It neglects denominator fluctuations and their correlations with $\mathcal{N}$, rather than assuming independence. Concentration alone does not provide a uniform finite-array error bound or determine the direction of the approximation error.

Second approximation: Substitution of the exact moments gives the intermediate expression I_{ratio} in Appendix A. Discard the correction $-1/N_{\mathrm{tx}}$ in its second normalized denominator factor and the corrections $-r + (1-r)/N_{\mathrm{tx}}$ to $N_{\mathrm{tx}} r$ in its numerator. The resulting mean-INR approximation is

$$I_{-} = \frac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}} \frac{\gamma_1^{\mathsf{U}} \left[N_{\mathrm{tx}} |\rho|^2 + \gamma_1^{\mathsf{U}} (1 - |\rho|^2) \right]}{(1 + \gamma_1^{\mathsf{U}}) \left[1 + \gamma_1^{\mathsf{U}} (1 - |\rho|^2) \right]}. \tag{31}$$

□

Here $r = |\rho|^2$ and $k = \mathcal{E}_{\mathsf{D}}/\mathcal{E}_{\mathsf{U}}$. The factor $1 - |\rho|^2$ follows from the exact cancellation and numerator moments. Neither I_{ratio} nor I_{-} is an exact finite-array mean or a guaranteed upper bound. The discarded terms can become leading terms when r or γ_1^{U} is small.

Figure 2 compares (31) with Monte Carlo for a half-wavelength ULA. Its response is $[\boldsymbol{a}(\theta)]_m = e^{j\pi m \cos\theta}$ for $m = 0, \ldots, N_{\mathrm{tx}} - 1$, with θ measured from the array axis. We set $\boldsymbol{h}_0 = \boldsymbol{a}(0^\circ)$ and $\boldsymbol{h}_1 = \sqrt{G_1}\boldsymbol{a}(15^\circ)$, assume the victim is detected, and retain the common receiver noise energy. With $k = 1/2000$ and fixed estimation-error variance, varying G_1 sweeps γ_1^{U} from -30 to 60 dB. The approximation captures the initial rise, the intermediate-SNR peaks for the smaller arrays, and the common plateau; the remaining deviations reflect the neglected finite-array terms. Since r also changes with the array size, a larger N_{tx} alone need not eliminate those deviations. The following subsection explains these regimes and establishes their exact limiting behavior.

Alternative derivation and interpretation: An alternative route approximates the leakage amplitude before taking its squared magnitude. It separates a deterministic contribution from a projected estimation-error term, evaluates their mean power, and approximates the beam normalization to recover the same expression (31). This gives a complementary interpretation of the two numerator terms and of the factor $1 - r$ as a projected-error variance. The complete derivation, including the discarded amplitude remainder and the normalization approximation, is given in Appendix B.

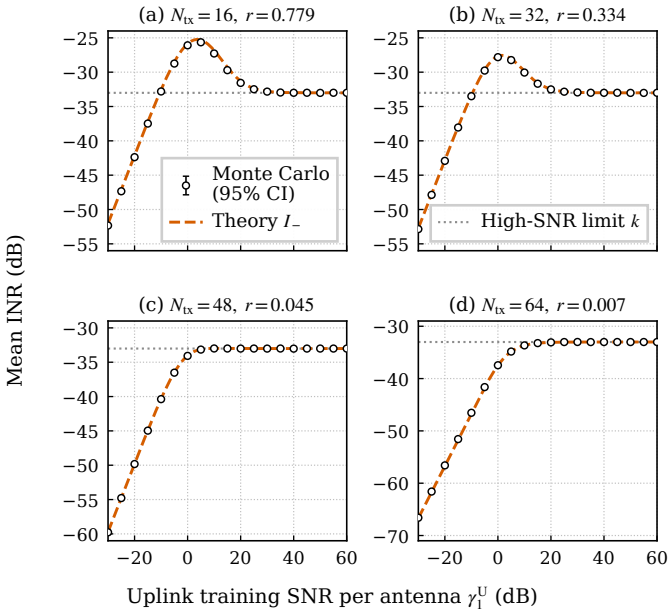


Fig. 2. Mean residual INR versus uplink training SNR per antenna for single-victim hard nulling with $N_{\mathrm{tx}} \in \{16, 32, 48, 64\}$. Markers average 5×10^5 independent Gaussian-error realizations of (28) in linear power before conversion to dB; error bars show pointwise approximate 95% confidence intervals for the mean. Dashed curves show I_- in (31); the dotted line marks $10\log_{10} k = -33.01$ dB. The channel correlation changes with the array size at fixed channel angles.

D. Low- and High-SNR Behavior

We now use (31) to characterize the initial growth and the eventual plateau of the mean INR, and check these predictions against exact limiting results from the original beam. To describe a sweep of channel strength, keep the transmit and training energies, N_0, N_{tx}, and the channel directions fixed, and vary G_1. Then k is constant and γ_1^{U} is proportional to G_1. Splitting the numerator of (31) gives $I_- = I_{\mathrm{bias}} + I_{\mathrm{noise}}$, where

$$\begin{cases} I_{\mathrm{bias}} = \dfrac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}} \dfrac{N_{\mathrm{tx}}|\rho|^2\gamma_1^{\mathsf{U}}}{(1+\gamma_1^{\mathsf{U}})[1+\gamma_1^{\mathsf{U}}(1-|\rho|^2)]}, \\ I_{\mathrm{noise}} = \dfrac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}} \dfrac{(\gamma_1^{\mathsf{U}})^2(1-|\rho|^2)}{(1+\gamma_1^{\mathsf{U}})[1+\gamma_1^{\mathsf{U}}(1-|\rho|^2)]}. \end{cases} \tag{32}$$

This decomposition follows directly from (31). Appendix B further shows that I_{bias} corresponds to the squared mean of the retained leakage amplitude and I_{noise} to its variance, after applying the same approximate normalization.

For $\gamma_1^{\mathsf{U}} \ll 1$, both denominator factors are close to one. The first contribution is then proportional to γ_1^{U}, whereas the second is proportional to $(\gamma_1^{\mathsf{U}})^2$. If $r > 0$ and $\gamma_1^{\mathsf{U}}(1-r) \ll N_{\mathrm{tx}}r$, the first contribution dominates and gives

$$\mathbb{E}[\mathsf{INR}_1] \approx \frac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}}\gamma_1^{\mathsf{U}}|\rho|^2 N_{\mathrm{tx}}. \tag{33}$$

□

Since $\gamma_1^{\mathsf{U}} = \mathcal{E}_{\mathsf{U}}G_1/N_0$, this is also $(\mathcal{E}_{\mathsf{D}}/N_0)N_{\mathrm{tx}}G_1 r$, the interference power of a beam along $\boldsymbol{u}$ with no victim-channel nulling. Thus, in this regime, nulling based on a noise-dominated estimate removes little of the original coupling for a large array. A tenfold increase in G_1 produces approximately a tenfold increase in mean INR, or a 10 dB rise. The factor $N_{\mathrm{tx}}r$ describes the array dependence at fixed channel correlation; in a particular array geometry, r can itself change with N_{tx}.

The finite-array slope can also be obtained directly from the original projected beam. At low training SNR, the estimated channel direction becomes isotropic. This yields the exact first-order expansion below for fixed $N_{\mathrm{tx}} \geq 2$ and fixed channel directions; the proof is given in Appendix C:

$$\mathbb{E}[\mathsf{INR}_1] = k\gamma_1^{\mathsf{U}}\left[(N_{\mathrm{tx}}-1)r + \frac{1-r}{N_{\mathrm{tx}}-1}\right] + o(\gamma_1^{\mathsf{U}}). \tag{34}$$

For $N_{\mathrm{tx}} \gg 1$ and $(N_{\mathrm{tx}}-1)^2 r \gg 1-r$, the coefficient in brackets is approximately $N_{\mathrm{tx}}r$, recovering (33). These conditions explain why a large array and a nonnegligible channel correlation are needed for the simplified slope. At $r = 0$, the exact leading term is instead $k\gamma_1^{\mathsf{U}}/(N_{\mathrm{tx}}-1)$; the simplified expression I_- misses this finite-array term and starts at order $(\gamma_1^{\mathsf{U}})^2$.

As the channel strength increases further, the same two terms in (32) exchange roles. For fixed $0 \leq r < 1$, the bias contribution decreases to zero and the noise contribution tends to k. Indeed, their high-SNR behavior is

$$\begin{cases} I_{\mathrm{bias}} = \dfrac{kN_{\mathrm{tx}}r}{\gamma_1^{\mathsf{U}}(1-r)}[1+o(1)] \longrightarrow 0, \\ I_{\mathrm{noise}} \longrightarrow k. \end{cases} \tag{35}$$

Thus the high-SNR part of the same approximation gives

$$\mathbb{E}[\mathsf{INR}_1] \approx \frac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}}, \qquad 0 \leq r < 1. \tag{36}$$

□

Sufficient conditions for this simplification of I_- are $\gamma_1^{\mathsf{U}} \gg 1$ and $\gamma_1^{\mathsf{U}}(1-r) \gg \max\{1, N_{\mathrm{tx}}r\}$. The second condition makes both the remaining denominator correction and I_{bias}/k small. Highly correlated channels can therefore require a much larger training SNR before the plateau is reached.

The plateau predicted by (36) is also an exact limit of the original model, independently of the moment approximations. For fixed $0 \leq r < 1$, as $G_1 \to \infty$ with the channel directions and training energy fixed, the estimated beam approaches the normalized nulling direction for the true victim channel. Combining this convergence with the exact error identity (22) gives the following result, proved in Appendix C:

$$\lim_{\gamma_1^{\mathsf{U}}\to\infty} \mathbb{E}[\mathsf{INR}_1] = \frac{\mathcal{E}_{\mathsf{D}}}{\mathcal{E}_{\mathsf{U}}} = k, \quad 0 \leq r < 1. \tag{37}$$

□

In physical terms, the limiting leakage contains one scalar component of estimation error, whose variance is σ^2 regardless of the array size. The plateau is independent of G_1 and N_{tx} at fixed energies. If the training energy itself is increased, k decreases, so the plateau level also decreases.

The plateau is an asymptotic value, not a uniform upper bound over all channel strengths. For example, direct subtraction in (31) gives

$$I_- - k = k\frac{[(N_{\mathrm{tx}}+1)r-2]\gamma_1^{\mathsf{U}} - 1}{(1+\gamma_1^{\mathsf{U}})[1+\gamma_1^{\mathsf{U}}(1-r)]}. \tag{38}$$

When $(N_{\mathrm{tx}}+1)r > 2$, the approximation exceeds k once $\gamma_1^{\mathsf{U}} > 1/[(N_{\mathrm{tx}}+1)r-2]$ and eventually approaches it from

above. Equation (37) does not imply $\mathbb{E}[\mathsf{INR}_1] \le k$ at finite SNR; the rigorous bound established here for all channel correlations remains (23). At $r = 1$, the desired direction aligns with the victim channel, so its projection onto the true-channel nullspace vanishes. The normalized limiting beam is undefined, so the plateau proof in Appendix C does not apply.

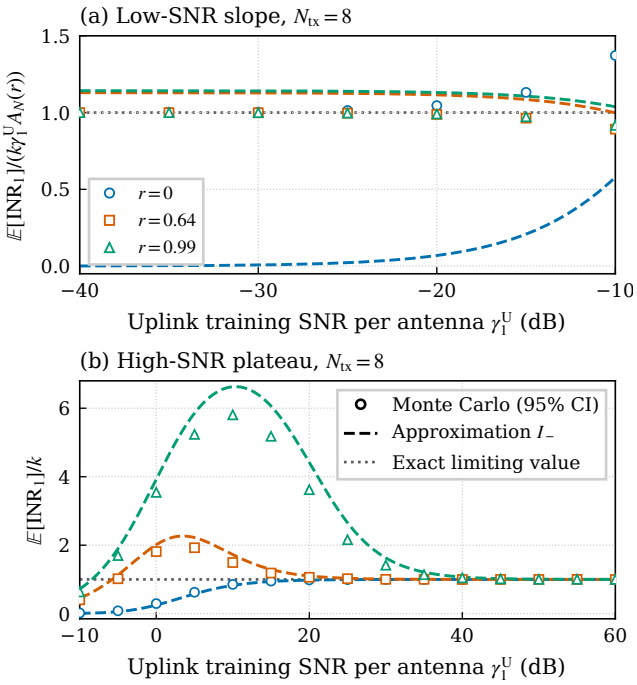


Fig. 3. Controlled finite-array validation for $N_{\mathrm{tx}} = 8$ and $r = |\rho|^2 \in \{0, 0.64, 0.99\}$. Each point averages 5×10^5 independent error realizations at fixed $k = 1/2000$ and estimation-error variance, without detection selection. Markers show Monte Carlo means with pointwise approximate 95% confidence intervals; dashed curves show I_- from (31) with the same normalization. (a) Mean INR divided by the exact low-SNR leading term $k\gamma_1^{\mathrm{U}}A_N(r)$, with limit one as $\gamma_1^{\mathrm{U}} \to 0$. (b) Mean INR divided by k, with limit one as $\gamma_1^{\mathrm{U}} \to \infty$ for each fixed $r < 1$. The dotted lines mark these exact limiting values, not finite-SNR bounds.

Numerical check of the limiting laws: Figure 3 isolates channel alignment by fixing $N = N_{\mathrm{tx}} = 8$ and independently setting $r \in \{0, 0.64, 0.99\}$ in the orthonormal representation (24). Write the exact low-SNR coefficient as $A_N(r) = (N-1)r + (1-r)/(N-1)$. In panel (a), the Monte Carlo mean divided by $k\gamma_1^{\mathrm{U}}A_N(r)$ tends to one, validating (34). The approximation has normalized limit $Nr/A_N(r)$, including zero at $r = 0$, illustrating the finite-array term omitted by (31). In panel (b), division by k gives the common unit limit in (37); the nearly parallel case has a larger intermediate-SNR excess and slower convergence. These checks distinguish the exact limiting laws from the finite-SNR accuracy of the mean-INR approximation.

Training-energy interpretation: For T known pilot symbols of energy $\mathcal{E}_{\mathrm{p}}$ each, $\mathcal{E}_{\mathrm{U}} = T\mathcal{E}_{\mathrm{p}}$. At fixed channel gain and per-symbol energy, increasing T changes both $\gamma_1^{\mathrm{U}} = T\mathcal{E}_{\mathrm{p}}G_1/N_0$ and $k = \mathcal{E}_{\mathrm{D}}/(T\mathcal{E}_{\mathrm{p}})$. Thus it differs from the fixed-k channel-gain sweeps above. Under the matched-filter model, records with the same total training energy have the same estimation-error distribution, so the plateau is a fixed-training-energy result.

IV. Practical Study and Performance Analysis

A. Site-Specific Simulation Setup

We consider a $15\,\mathrm{km} \times 15\,\mathrm{km}$ scene centered on CableLabs in Louisville, Colorado (39.9570131°N, 105.1603427°W), using the Sionna ray-tracing platform [16]. Buildings and roads are retained from the site-specific Blender model, with flat ground in this experiment. Fig. 4 shows the saved deployment on a georeferenced aerial basemap. Local east and north coordinates share the CableLabs origin of the scene's spherical transverse Mercator projection. The basemap provides geographic context; propagation is computed from the three-dimensional scene geometry.

The terrestrial network contains seven sites with a $5\,\mathrm{km}$ intersite distance and three sectors per site. One TN-UE is scheduled in each of the 21 hexagonal cells. TN-UE locations satisfy a $3\,\mathrm{dB}$ serving-sector channel-power advantage over every competing sector and a $35\,\mathrm{m}$ minimum site separation. This realization contains 14 outdoor and seven indoor TN-UEs. The 100 NTN-UEs comprise 50 rooftop and 50 outdoor terminals, sampled across the full square scene, including locations outside the terrestrial cell union. The central region $|x| < 250\,\mathrm{m}$, $|y| < 800\,\mathrm{m}$ is excluded from NTN placement. Table I lists the parameters of this illustrative realization.

TABLE I
Site-Specific Radio-Map Parameters.

Radio Parameters	**Values**
Carrier Frequency and Bandwidth	9.999 GHz and 100 MHz
Noise Power Spectral Density	−174 dBm/Hz
Downlink Transmit Power	30 dBm per active sector
Receiver Noise Figure	3 dB (NTN-UE), 7 dB (TN-UE)
Maximum RT Interaction Depth	3; specular reflection and refraction
NTN Beam-Design Channels	Perfect LoS CSI; $\lambda = 10^{12}$
Radio-Map Grid Spacing	10 m
Antenna Type and Configuration	
Base Station (BS)	8×8 URA, Directional [17], 3 sectors
Handheld (TN-UE)	Single vertically polarized omnidirectional element
VSAT (NTN-UE)	60 cm diameter, Directional
Antenna Heights	40 m (BS), 1.88 m (outdoor NTN), 1.2 m above roof (NTN)
Scene Configuration	
Geographical Location	15 km × 15 km, CableLabs
Radio Material	Brick walls, concrete roofs, medium-dry ground, asphalt roads
Satellite Look Direction	Azimuth 317.58°, elevation 82.71°
NTN-UE Deployment	50 rooftop, 50 outdoor
TN-UE Deployment	21 total; one per sector
TN-BS Deployment	7 sites, 5 km intersite distance, 5° downtilt

B. Spatial Interference Suppression

Fig. 5 compares the aggregate NTN downlink INR before and after nulling for the same deployment. All 21 sectors transmit independent data streams at fixed power. At map location $\boldsymbol{r}$, let $\boldsymbol{h}_b(\boldsymbol{r})$ denote the effective channel from sector b to the single-antenna NTN receiver. The network INR is

$$\mathsf{INR}^{\mathrm{net}}(\boldsymbol{r}) = \frac{P_{\mathrm{BS}}}{N_{\mathrm{NTN}}} \sum_{b=1}^{21} \left| \boldsymbol{w}_b^H \boldsymbol{h}_b(\boldsymbol{r}) \right|^2, \tag{39}$$

where P_{BS} is the power per sector and N_{NTN} is the NTN receiver noise power over the simulated bandwidth. Complex path contributions are summed coherently within each link; powers are then summed across independently transmitting sectors.

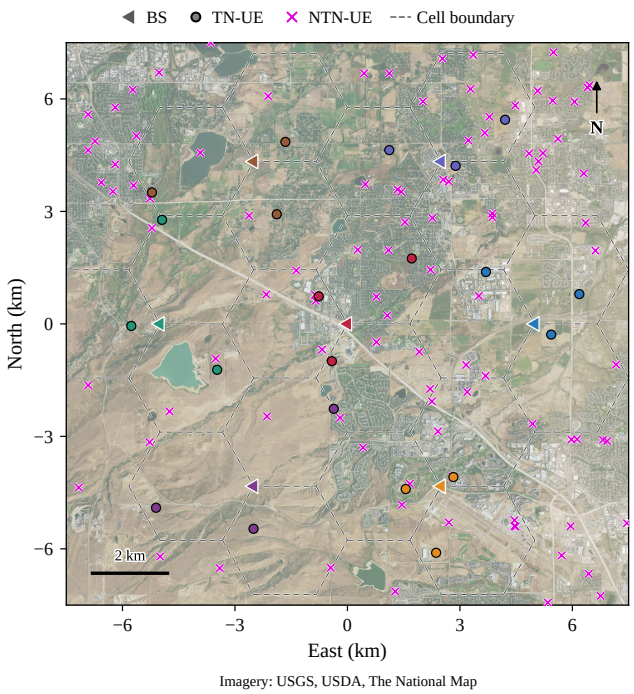
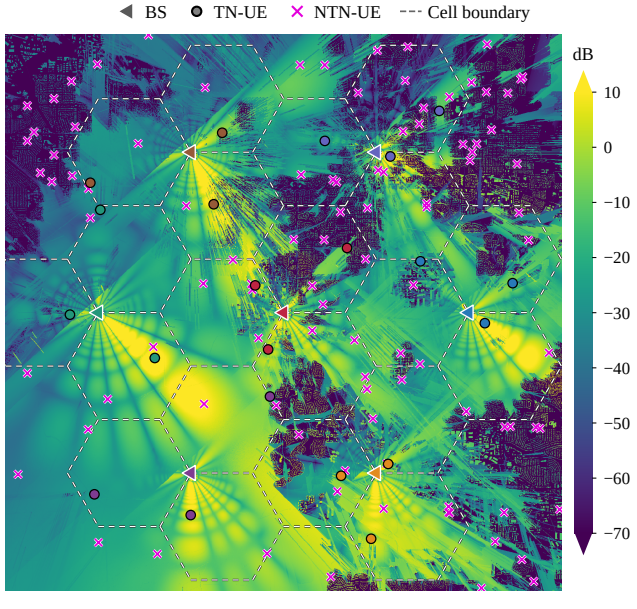
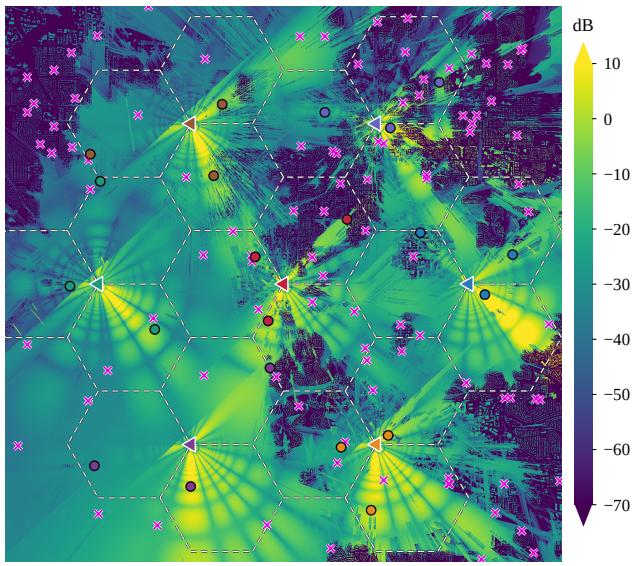


Fig. 4. Site-specific deployment centered on CableLabs. Triangles, circles, and magenta crosses denote seven BS sites, 21 scheduled TN-UEs, and 100 NTN-UEs, respectively. BS and TN-UE colors identify the serving site; dashed lines delimit the 21 cells, clipped to the scene boundary. The aerial basemap is reconstructed from USGS, USDA, The National Map imagery using the saved scene georeference.



(a) Before nulling



(b) After nulling

Fig. 5. Aggregate NTN downlink INR (a) before nulling and (b) after perfect-LoS-CSI nulling with $\lambda = 10^{12}$ for the deployment in Fig. 4. Both panels use the same $10\,\mathrm{m}$ grid, depth-three multipath evaluation, and color limits over the full square scene. Triangles, circles, and crosses denote BS sites, TN-UEs, and NTN-UEs; dashed lines delimit the cells.

The baseline beams maximize the desired TN-link gain. The nulling beams use perfect LoS NTN channels and penalty weight $\lambda = 10^{12}$, with the serving TN channel normalized in the beam-design objective. TN beam-design channels include multipath. Both maps evaluate the saved beams with LoS, specular reflection, and refraction, with a maximum interaction depth of three. Thus the perfect LoS design does not imply perfect knowledge of the multipath evaluation channel. This experiment isolates spatial suppression under that design/evaluation mismatch; it does not evaluate channel-estimation noise.

Map receivers are placed on a $10\,\mathrm{m}$ grid, at $1.2\,\mathrm{m}$ above rooftops or $1.88\,\mathrm{m}$ above outdoor ground, with the same satellite look direction as the deployed NTN terminals. The two panels share the color scale and show the full $15\,\mathrm{km}\times15\,\mathrm{km}$ square scene. They illustrate the spatial change in interference for one fixed realization, including residual multipath leakage after LoS-based nulling.

C. CDF Performance with Noisy Channel Estimation

We next perform ten independent deployment realizations in the same scene, retaining the BS positions, 8×8 arrays, $40\,\mathrm{m}$ BS height, and 5° downtilt. Each realization contains one admitted TN-UE per sector and 100 NTN-UEs sampled using the same full-scene, 50-rooftop/50-outdoor rule. The TN indoor/outdoor type is drawn with equal probability and retained during admission retries. Satellite azimuth and elevation are sampled uniformly over $[0, 360)^\circ$ and $[25, 90]^\circ$, respectively. The first realization uses the exact deployment and satellite direction shown in Fig. 4. Pooling the realizations gives 1,000 NTN INR samples and 210 TN SINR samples per method.

For this CDF experiment, channel estimation, beam design, and evaluation all use the effective narrowband multipath channels, with up to three interactions and 10^5 ray samples per source. We compare no nulling with true-channel and noisy-estimated-channel nulling for $\lambda \in \{10^{11}, 10^{12}, 10^{13}\}$. The serving TN CSI is exact and normalized in the beam-design objective. The true-channel benchmark includes ev-

ery NTN channel in its interference penalty. For estimated-channel nulling, independent circular complex Gaussian noise is added to the effective NTN channel coefficients. A $30\,\mathrm{dBm}$, $20\,\mu\mathrm{s}$ pilot, $-174\,\mathrm{dBm/Hz}$ thermal noise PSD, and $2\,\mathrm{dB}$ BS noise figure give a per-coefficient estimation-error variance of 3.155×10^{-16}. Only estimated links exceeding the noise-calibrated energy-detection threshold, with per-link false-alarm probability 10^{-8}, enter the estimated interference penalty. The same true channels and one estimation-noise draw are shared by all three weights within each realization. Evaluation always uses the true channels, including links missed by detection.

Fig. 6 shows the aggregate NTN INR. Increasing λ reduces interference: the empirical fraction above $-6\,\mathrm{dB}$ falls from 34.2% without nulling to 14.1% and 0.6% for estimated-channel nulling with $\lambda = 10^{11}$ and 10^{12}, respectively; none of the 1,000 samples exceeds this threshold at $\lambda = 10^{13}$. At $\lambda = 10^{13}$, the median INR is approximately $-31.7\,\mathrm{dB}$ with true CSI and $-27.7\,\mathrm{dB}$ with noisy CSI, compared with $-12.7\,\mathrm{dB}$ without nulling. These empirical distributions retain the 84 samples with no traced coupling from any sector. The displayed range is $-60\,\mathrm{dB}$ to $20\,\mathrm{dB}$; all samples outside this range, including zero-power samples, remain in the CDF denominator.

Fig. 7 quantifies the corresponding TN cost. The SINR denominator includes receiver noise and the sum of interference powers from all 20 other sectors, including the two other sectors at the serving site. For estimated-channel nulling with $\lambda = 10^{13}$, the median SINR decreases from approximately $20.2\,\mathrm{dB}$ to $17.0\,\mathrm{dB}$, while its tenth percentile decreases from $10.2\,\mathrm{dB}$ to $-3.4\,\mathrm{dB}$. The corresponding true-CSI median is $16.7\,\mathrm{dB}$. Thus the strongest penalty gives substantial NTN interference suppression but also reduces the lower tail of TN performance; the smaller weights provide intermediate tradeoffs.

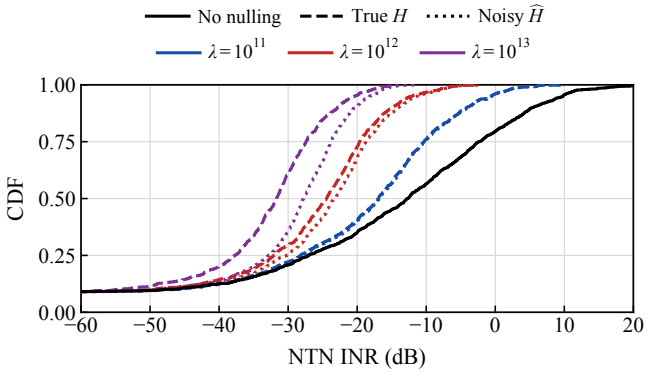


Fig. 6. NTN INR CDF over ten realizations (1,000 samples). Black solid: no nulling; colored dashed: true-channel nulling; colored dotted: noisy-estimated-channel nulling with detection. Colors identify the three penalty weights. Interference power is summed over all 21 sectors, including links missed by detection. The horizontal axis spans $-60\,\mathrm{dB}$ to $20\,\mathrm{dB}$; samples outside the displayed range are retained in the CDF.

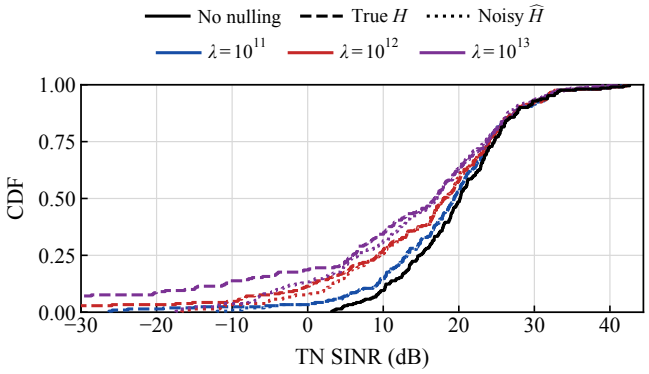


Fig. 7. TN SINR CDF for the same ten realizations (210 samples), with the same line styles and colors as Fig. 6. All methods use the same UE drops and true evaluation channels. The denominator includes noise and interference from every nonserving sector. The displayed range starts at $-30\,\mathrm{dB}$, with all samples retained in the CDF.

Appendix A
Derivation of the Exact Leakage Representation and Mean-INR Approximation

This appendix supplies the algebra and moment calculations underlying Sections III-B and III-C. The leakage representation and its moments are exact. Denominator replacement and finite-array truncation then yield (31).

A. Coordinate Construction and Exact Cancellation

The first direction in (24) is

$$\boldsymbol{e}_1 = \frac{\boldsymbol{h}_1}{\sqrt{H}}. \tag{40}$$

The phase convention in (16) gives $\boldsymbol{e}_1^H \boldsymbol{u} = \sqrt{r}$. For $r < 1$, the second direction is

$$\boldsymbol{e}_2 = \frac{\boldsymbol{u} - \sqrt{r}\,\boldsymbol{e}_1}{\sqrt{1-r}}. \tag{41}$$

These vectors are orthonormal. For $r = 1$, any unit vector orthogonal to $\boldsymbol{e}_1$ suffices because its coefficient in (24) is zero.

With the error coordinates defined in Section III-B, $\boldsymbol{\xi} = (\boldsymbol{I} - \boldsymbol{e}_1\boldsymbol{e}_1^H)\boldsymbol{v}_1$. Isotropic Gaussian noise has independent components on orthogonal subspaces, so ζ is independent of $\boldsymbol{\xi}$ and $c = \sqrt{H} + \zeta$ is independent of (x, Q). The projected energy satisfies $U = \|(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^H)\widehat{\boldsymbol{h}}_1\|^2$. Using (18) and (26) therefore gives

$$p = \frac{U}{T_{\mathrm{h}}}. \tag{42}$$

Inserting the projector into the leakage amplitude yields

$$\begin{aligned} F &= \boldsymbol{u}^H \boldsymbol{P}_{\widehat{h}_1}^{\perp} \boldsymbol{h}_1 \\ &= \boldsymbol{u}^H \left(\boldsymbol{I} - \frac{\widehat{\boldsymbol{h}}_1 \widehat{\boldsymbol{h}}_1^H}{T_{\mathrm{h}}} \right) \boldsymbol{h}_1 \\ &= \boldsymbol{u}^H \boldsymbol{h}_1 - \frac{(\boldsymbol{u}^H \widehat{\boldsymbol{h}}_1)(\widehat{\boldsymbol{h}}_1^H \boldsymbol{h}_1)}{T_{\mathrm{h}}}. \end{aligned} \tag{43}$$

The three scalar inner products can be evaluated separately using (24) and $\widehat{\boldsymbol{h}}_1 = c\boldsymbol{e}_1 + \boldsymbol{\xi}$. Since $\boldsymbol{e}_1^H\boldsymbol{e}_1 = 1$, $\boldsymbol{e}_2^H\boldsymbol{e}_1 = 0$, $\boldsymbol{e}_1^H\boldsymbol{\xi} = 0$, and $\boldsymbol{e}_2^H\boldsymbol{\xi} = x$, they are

$$\begin{aligned} \boldsymbol{u}^H\boldsymbol{h}_1 &= (\sqrt{r}\,\boldsymbol{e}_1^H + \sqrt{1-r}\,\boldsymbol{e}_2^H)\sqrt{H}\,\boldsymbol{e}_1 \\ &= \sqrt{Hr}, \\ \boldsymbol{u}^H\widehat{\boldsymbol{h}}_1 &= (\sqrt{r}\,\boldsymbol{e}_1^H + \sqrt{1-r}\,\boldsymbol{e}_2^H)(c\boldsymbol{e}_1 + \boldsymbol{\xi}) \\ &= \sqrt{r}\,c + \sqrt{1-r}\,x, \\ \widehat{\boldsymbol{h}}_1^H\boldsymbol{h}_1 &= (c^*\boldsymbol{e}_1^H + \boldsymbol{\xi}^H)\sqrt{H}\,\boldsymbol{e}_1 \\ &= c^*\sqrt{H}. \end{aligned}$$

In the last inner product, the Hermitian transpose conjugates c; the coefficient x in the second inner product is not conjugated. Putting these terms over the common denominator and using $T_{\mathrm{h}} = |c|^2 + Q$ exposes the cancellation:

$$\begin{aligned} F &= \sqrt{Hr} - \frac{(\sqrt{r}\,c + \sqrt{1-r}\,x)c^*\sqrt{H}}{T_{\mathrm{h}}} \\ &= \frac{\sqrt{H}}{T_{\mathrm{h}}}\left[\sqrt{r}\,T_{\mathrm{h}} - \sqrt{r}\,|c|^2 - \sqrt{1-r}\,xc^*\right] \\ &= \frac{\sqrt{H}}{T_{\mathrm{h}}}\left[\sqrt{r}\,(|c|^2 + Q) - \sqrt{r}\,|c|^2 - \sqrt{1-r}\,xc^*\right] \\ &= \frac{\sqrt{H}}{T_{\mathrm{h}}}\left[\sqrt{r}\,Q - \sqrt{1-r}\,xc^*\right]. \end{aligned}$$

The common term is $|c|^2 = H + 2\sqrt{H}\,\mathrm{Re}(\zeta) + |\zeta|^2$; its channel-energy and error-dependent parts cancel together. Finally, $|\widehat{\boldsymbol{w}}^H\boldsymbol{h}_1|^2 = |F|^2/p$ and (42) give

$$\begin{aligned} \frac{|F|^2}{p} &= \frac{H|\sqrt{r}\,Q - \sqrt{1-r}\,xc^*|^2}{T_{\mathrm{h}}^2}\frac{T_{\mathrm{h}}}{U} \\ &= \frac{H|\sqrt{r}\,Q - \sqrt{1-r}\,xc^*|^2}{T_{\mathrm{h}}U}, \end{aligned}$$

which verifies (28).

An exact representation for direct numerical averaging: The error energy outside both selected directions is

$$R_{\perp} = \|(\boldsymbol{I} - \boldsymbol{e}_1\boldsymbol{e}_1^H - \boldsymbol{e}_2\boldsymbol{e}_2^H)\boldsymbol{v}_1\|^2. \tag{44}$$

It is independent of ζ and x. For $N_{\mathrm{tx}} > 2$, $R_{\perp}/\sigma^2 \sim \mathrm{Gamma}(N_{\mathrm{tx}} - 2, 1)$, with shape $N_{\mathrm{tx}} - 2$ and scale one. For $N_{\mathrm{tx}} = 2$ there are no remaining directions, and $R_{\perp} = 0$ almost surely. The energies become

$$\begin{cases} Q = |x|^2 + R_{\perp}, \\ U = |\sqrt{1-r}\,c - \sqrt{r}\,x|^2 + R_{\perp}. \end{cases} \tag{45}$$

Consequently, one can sample the independent quantities ζ, x, and $R_{\perp}$, form c, Q, T_{h}, and U, and average (28). This representation is valid for every $N_{\mathrm{tx}} \geq 2$, including small arrays.

B. Exact Numerator Moments and Denominator Means

Consider the numerator $\mathcal{N}$ defined in (29). Expanding its squared modulus gives

$$\begin{aligned} \mathcal{N} = H\big[&rQ^2 + (1-r)|x|^2|c|^2 \\ &- 2\sqrt{r(1-r)}\,\mathrm{Re}(Qx^*c)\big]. \end{aligned} \tag{46}$$

The four moments needed here are

$$\begin{aligned} \mathbb{E}[Q^2] &= N_{\mathrm{tx}}(N_{\mathrm{tx}}-1)\sigma^4, & \mathbb{E}[|x|^2] &= \sigma^2, \\ \mathbb{E}[|c|^2] &= H + \sigma^2, & \mathbb{E}[Qx^*c] &= 0. \end{aligned} \tag{47}$$

For the first identity, Q/σ^2 is the sum of $N_{\mathrm{tx}} - 1$ independent unit-mean exponential random variables. Its mean and variance both equal $N_{\mathrm{tx}} - 1$, giving the stated second moment. The next two moments follow from $x \sim \mathcal{CN}(0, \sigma^2)$ and $c = \sqrt{H} + \zeta$. For the last moment, circular symmetry gives $\mathbb{E}[Qx^*] = 0$, and c is independent of (Q, x). Finally, independence of x and c gives $\mathbb{E}[|x|^2|c|^2] = \sigma^2(H + \sigma^2)$. Taking the expectation of the expanded numerator therefore gives

$$\begin{aligned} \mathbb{E}\mathcal{N} &= H\left[rN_{\mathrm{tx}}(N_{\mathrm{tx}}-1)\sigma^4 + (1-r)\sigma^2(H+\sigma^2)\right] \\ &= H\sigma^4\left[rN_{\mathrm{tx}}(N_{\mathrm{tx}}-1) + (1-r)(N_{\mathrm{tx}}\gamma_1^{\mathsf{U}} + 1)\right]. \end{aligned} \tag{48}$$

Exact denominator means: The total energy T_{h} contains channel energy H and mean noise energy $N_{\mathrm{tx}}\sigma^2$. The projection defining U retains channel energy $H(1-r)$ and noise in $N_{\mathrm{tx}} - 1$ dimensions. Consequently,

$$\begin{cases} \mathbb{E}[T_{\mathrm{h}}] = N_{\mathrm{tx}}\sigma^2(1 + \gamma_1^{\mathsf{U}}), \\ \mathbb{E}[U] = \sigma^2\left[N_{\mathrm{tx}} - 1 + N_{\mathrm{tx}}\gamma_1^{\mathsf{U}}(1-r)\right]. \end{cases} \tag{49}$$

All moments obtained so far are exact; the remaining difficulty is the expectation of a ratio of dependent random quantities.

C. Denominator Concentration and Approximation

To assess the denominator fluctuations neglected in (30), the Gaussian error model gives

$$\begin{aligned} \frac{\mathrm{Var}(T_{\mathrm{h}})}{(\mathbb{E}T_{\mathrm{h}})^2} &= \frac{1 + 2\gamma_1^{\mathsf{U}}}{N_{\mathrm{tx}}(1+\gamma_1^{\mathsf{U}})^2}, \\ \frac{\mathrm{Var}(U)}{(\mathbb{E}U)^2} &= \frac{N_{\mathrm{tx}} - 1 + 2N_{\mathrm{tx}}\gamma_1^{\mathsf{U}}(1-r)}{\left[N_{\mathrm{tx}} - 1 + N_{\mathrm{tx}}\gamma_1^{\mathsf{U}}(1-r)\right]^2}. \end{aligned} \tag{50}$$

For $N_{\mathrm{tx}} \to \infty$ with $\gamma_1^{\mathsf{U}} > 0$ and r fixed, both expressions are $O(N_{\mathrm{tx}}^{-1})$. Thus each denominator norm divided by its mean converges to one in mean square; its relative standard deviation is $O(N_{\mathrm{tx}}^{-1/2})$. This motivates replacing both norms by their means *after* the exact numerator cancellation in (28). Equation (30) neglects the remaining denominator fluctuations and their correlations with $\mathcal{N}$; it does not assume that $\mathcal{N}$, T_{h}, and U are independent. The concentration argument alone is

not a uniform finite-array error bound and gives no guaranteed direction for the approximation error. Multiply (30) by $\mathcal{E}_{\text{D}}/N_0$ as in (21). Substitution of the exact moments then gives $\mathbb{E}[\mathsf{INR}_1] \approx I_{\text{ratio}}$, where

$$I_{\text{ratio}} = k\gamma_1^{\text{U}} \frac{r(N_{\text{tx}}-1) + (1-r)(\gamma_1^{\text{U}} + 1/N_{\text{tx}})}{(1+\gamma_1^{\text{U}})\,[1 - 1/N_{\text{tx}} + \gamma_1^{\text{U}}(1-r)]}. \tag{51}$$

The second approximation drops $-1/N_{\text{tx}}$ in the second denominator factor and $-r + (1-r)/N_{\text{tx}}$ in the numerator, leaving $N_{\text{tx}}r + \gamma_1^{\text{U}}(1-r)$ there. This yields (31). The exact moments do not remove the approximation in (30); both I_{ratio} and I_{-} remain approximations to the finite-array mean.

Appendix B
Alternative Derivation of the Mean Residual INR

Complementing the moment calculation in Appendix A, this appendix derives (31) by approximating the leakage amplitude before squaring and averaging. It separates a deterministic contribution from a projected estimation-error term and identifies the discarded remainder explicitly. The resulting mean and variance also explain the two terms in (32).

Define the scalar channel and error projections

$$\begin{aligned} C &= \boldsymbol{u}^H \boldsymbol{h}_1, \quad d = \boldsymbol{u}^H \boldsymbol{v}_1, \\ t &= \boldsymbol{h}_1^H \boldsymbol{v}_1, \quad q = \|\boldsymbol{v}_1\|^2. \end{aligned} \tag{52}$$

Here $C = \sqrt{Hr}$ is the fixed component of the true victim channel along the desired direction. The variables d and t are error projections onto $\boldsymbol{u}$ and $\boldsymbol{h}_1$, while q is the total error energy. Recall that F is the amplitude defined in (20), so $|\widehat{\boldsymbol{w}}^H \boldsymbol{h}_1|^2 = |F|^2/p$; it is distinct from the power numerator $\mathcal{N}$ defined in (29). Using $\widehat{\boldsymbol{h}}_1 = \boldsymbol{h}_1 + \boldsymbol{v}_1$, expand the estimated-channel energy as

$$\begin{aligned} T_{\text{h}} &= \|\widehat{\boldsymbol{h}}_1\|^2 \\ &= (\boldsymbol{h}_1 + \boldsymbol{v}_1)^H (\boldsymbol{h}_1 + \boldsymbol{v}_1) \\ &= \boldsymbol{h}_1^H \boldsymbol{h}_1 + \boldsymbol{h}_1^H \boldsymbol{v}_1 + \boldsymbol{v}_1^H \boldsymbol{h}_1 + \boldsymbol{v}_1^H \boldsymbol{v}_1 \\ &= H + t + t^* + q. \end{aligned} \tag{53}$$

Here $\boldsymbol{v}_1^H \boldsymbol{h}_1 = (\boldsymbol{h}_1^H \boldsymbol{v}_1)^* = t^*$, so the two cross terms sum to $2\,\mathrm{Re}(t)$. The inner products in the projector expansion (43) become

$$\boldsymbol{u}^H \widehat{\boldsymbol{h}}_1 = C + d, \qquad \widehat{\boldsymbol{h}}_1^H \boldsymbol{h}_1 = H + t^*.$$

Substituting these identities and (53) into (43) gives the first line below. For the second line, the error identity (22) gives $F = -\boldsymbol{u}^H \boldsymbol{P}_{\widehat{\boldsymbol{h}}_1}^{\perp} \boldsymbol{v}_1$; expanding this projector and using $\widehat{\boldsymbol{h}}_1^H \boldsymbol{v}_1 = t + q$ gives the alternative expression. Thus the same exact amplitude has two forms:

$$\begin{aligned} F &= C - \frac{(C+d)(H+t^*)}{H + t + t^* + q} \\ &= -d + \frac{(C+d)(t+q)}{H + t + t^* + q}. \end{aligned} \tag{54}$$

To separate the mean denominator from its fluctuation, define

$$\begin{cases} M = \mathbb{E}[T_{\text{h}}] = H + N_{\text{tx}}\sigma^2, \\ \alpha = H/M, \\ \delta = T_{\text{h}} - M = t + t^* + q - N_{\text{tx}}\sigma^2. \end{cases} \tag{55}$$

Thus α is the fraction of the mean estimated-channel energy contributed by the true channel, and δ is the random energy fluctuation. Define the projector orthogonal to the true victim channel by

$$\boldsymbol{P}_{h_1}^{\perp} = \boldsymbol{I} - \frac{\boldsymbol{h}_1 \boldsymbol{h}_1^H}{H}. \tag{56}$$

Using this projector, we can express the retained amplitude in terms of projected estimation error. Direct rearrangement of (54) gives

$$F = F_{\text{red}} + \mathcal{E}, \tag{57}$$

where the retained amplitude is

$$\begin{aligned} F_{\text{red}} &= C(1-\alpha) - \alpha d + \frac{Ct}{M} \\ &= C(1-\alpha) - \alpha \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{v}_1, \end{aligned} \tag{58}$$

The remainder collects exactly the terms omitted from this amplitude:

$$\begin{aligned} \mathcal{E} = \frac{dt}{M} &+ \frac{(C+d)(q - N_{\text{tx}}\sigma^2)}{M} \\ &- \frac{(C+d)(t+q)\delta}{M(M+\delta)}. \end{aligned} \tag{59}$$

The decomposition including $\mathcal{E}$ is exact. The approximation consists of discarding this remainder and replacing p by $1 - \alpha r$. To compute the moments of the retained amplitude, note that $\boldsymbol{u}$ and $\boldsymbol{P}_{h_1}^{\perp}$ depend only on the fixed true channels. Since $\mathbb{E}[\boldsymbol{v}_1] = \boldsymbol{0}$, the projected error has zero mean:

$$\mathbb{E}[\boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{v}_1] = \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \mathbb{E}[\boldsymbol{v}_1] = 0.$$

Using the error covariance $\mathbb{E}[\boldsymbol{v}_1 \boldsymbol{v}_1^H] = \sigma^2 \boldsymbol{I}$, its mean squared magnitude is

$$\begin{aligned} &\mathbb{E}\big[|\boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{v}_1|^2\big] \\ &\quad = \mathbb{E}\big[\boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{v}_1 \boldsymbol{v}_1^H (\boldsymbol{P}_{h_1}^{\perp})^H \boldsymbol{u}\big] \\ &\quad = \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \mathbb{E}[\boldsymbol{v}_1 \boldsymbol{v}_1^H] (\boldsymbol{P}_{h_1}^{\perp})^H \boldsymbol{u} \\ &\quad = \sigma^2 \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} (\boldsymbol{P}_{h_1}^{\perp})^H \boldsymbol{u} \\ &\quad = \sigma^2 \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{u}. \end{aligned}$$

The last equality uses the Hermitian and idempotent properties of an orthogonal projector:

$$(\boldsymbol{P}_{h_1}^{\perp})^H = \boldsymbol{P}_{h_1}^{\perp}, \qquad (\boldsymbol{P}_{h_1}^{\perp})^2 = \boldsymbol{P}_{h_1}^{\perp}.$$

Next, expand the remaining quadratic form using (56), $\|\boldsymbol{u}\| = 1$, and $|\boldsymbol{u}^H \boldsymbol{h}_1|^2 = Hr$:

$$\begin{aligned} \boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{u} &= \boldsymbol{u}^H \left(\boldsymbol{I} - \frac{\boldsymbol{h}_1 \boldsymbol{h}_1^H}{H} \right) \boldsymbol{u} \\ &= 1 - \frac{|\boldsymbol{u}^H \boldsymbol{h}_1|^2}{H} \\ &= 1 - \frac{Hr}{H} = 1 - r. \end{aligned}$$

Substituting this identity into the moment calculation gives

$$\mathbb{E}\big[|\boldsymbol{u}^H \boldsymbol{P}_{h_1}^{\perp} \boldsymbol{v}_1|^2\big] = \sigma^2 (1-r). \tag{60}$$

In (58), $C(1-\alpha)$ is deterministic and the projected-error term is zero-mean. Expanding the squared magnitude before taking its expectation gives

$$|F_{\mathrm{red}}|^2 = |C|^2(1-\alpha)^2 + \alpha^2|\boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{v}_1|^2 - 2\alpha(1-\alpha)\,\mathrm{Re}\big(C^*\boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{v}_1\big).$$

The cross term vanishes in expectation because

$$\mathbb{E}\big[\mathrm{Re}\big(C^*\boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{v}_1\big)\big] = \mathrm{Re}\big(C^*\mathbb{E}[\boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{v}_1]\big) = 0.$$

Consequently, the deterministic and random contributions add in power:

$$\mathbb{E}[|F_{\mathrm{red}}|^2] = |C|^2(1-\alpha)^2 + \alpha^2\mathbb{E}\big[|\boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{v}_1|^2\big].$$

Using $|C|^2 = Hr$ and (60) yields

$$\mathbb{E}[|F_{\mathrm{red}}|^2] = Hr(1-\alpha)^2 + \alpha^2\sigma^2(1-r). \tag{61}$$

To connect this mean squared amplitude to the mean INR, use the exact relation (21) and the amplitude approximation specified after (59). The remaining normalization approximation can be obtained from (18) and (52):

$$p = 1 - \frac{|C+d|^2}{T_{\mathrm{h}}}.$$

Since $d = \boldsymbol{u}^H\boldsymbol{v}_1$, the error model and $\|\boldsymbol{u}\| = 1$ give

$$\mathbb{E}[d] = 0, \qquad \mathbb{E}[|d|^2] = \boldsymbol{u}^H(\sigma^2\boldsymbol{I})\boldsymbol{u} = \sigma^2.$$

Consequently,

$$\mathbb{E}[|C+d|^2] = |C|^2 + 2\,\mathrm{Re}(C^*\mathbb{E}[d]) + \mathbb{E}[|d|^2] = Hr + \sigma^2.$$

Replacing the numerator and denominator energies in p by their respective means, with $M = \mathbb{E}[T_{\mathrm{h}}]$ as defined in (55), gives

$$p \approx 1 - \frac{Hr+\sigma^2}{M} = 1 - \alpha r - \frac{\sigma^2}{M}.$$

The last term is the finite-array correction

$$\frac{\sigma^2}{M} = \frac{\sigma^2}{H + N_{\mathrm{tx}}\sigma^2} = \frac{1}{N_{\mathrm{tx}}(1+\gamma_1^{\mathrm{U}})}.$$

Dropping this correction gives the stated approximation $p \approx 1 - \alpha r$. This substitutes a deterministic value for the random normalization; it is not an exact evaluation of $\mathbb{E}[p]$. With this substitution and the retained amplitude, the denominator can be taken outside the expectation in (21). Inserting (61) then gives

$$\begin{aligned}\mathbb{E}[\mathsf{INR}_1] &\approx \frac{\mathcal{E}_{\mathrm{D}}}{N_0}\frac{\mathbb{E}[|F_{\mathrm{red}}|^2]}{1-\alpha r} \\ &= \frac{\mathcal{E}_{\mathrm{D}}}{N_0}\frac{Hr(1-\alpha)^2 + \alpha^2\sigma^2(1-r)}{1-\alpha r} \\ &= k\,\frac{\gamma_1^{\mathrm{U}}\,[N_{\mathrm{tx}}r + \gamma_1^{\mathrm{U}}(1-r)]}{(1+\gamma_1^{\mathrm{U}})\,[1+\gamma_1^{\mathrm{U}}(1-r)]} = I_-.\end{aligned} \tag{62}$$

The last line uses $H = N_{\mathrm{tx}}\sigma^2\gamma_1^{\mathrm{U}}$, $\alpha = \gamma_1^{\mathrm{U}}/(1+\gamma_1^{\mathrm{U}})$, and $(\mathcal{E}_{\mathrm{D}}/N_0)\sigma^2 = k$, and is identical to (31). Thus the neglected terms are explicitly identifiable. They cannot in general be set to zero merely because $\boldsymbol{v}_1$ has zero mean.

APPENDIX C
PROOFS OF THE EXACT LOW- AND HIGH-SNR LIMITS

The following proofs use the original projected beam and the exact estimation-error model. They do not use the mean-INR approximation in (31) or the retained-amplitude approximation in Appendix B. As in Section III-D, the transmit and training energies, noise energy, antenna count, and channel directions are fixed while G_1 varies.

A. Low-SNR Expansion

We prove the first-order expansion in (34). As $\gamma_1^{\mathrm{U}} \to 0$ at fixed $N_{\mathrm{tx}} \geq 2$ and fixed channel directions, the estimated channel direction becomes isotropic. Consequently, $\mathbb{E}[p] \to 1 - 1/N_{\mathrm{tx}}$. The normalized projected beam in (19) satisfies $\boldsymbol{u}^H\widehat{\boldsymbol{w}} = \sqrt{p}$, so its mean energy along $\boldsymbol{u}$ tends to $(N_{\mathrm{tx}}-1)/N_{\mathrm{tx}}$. Its total energy is one, and rotational symmetry spreads the remaining mean energy $1/N_{\mathrm{tx}}$ equally over the $N_{\mathrm{tx}}-1$ orthogonal directions, with zero cross terms in this decomposition. Thus

$$\lim_{\gamma_1^{\mathrm{U}}\to 0}\mathbb{E}[\widehat{\boldsymbol{w}}\widehat{\boldsymbol{w}}^H] = \frac{N_{\mathrm{tx}}-1}{N_{\mathrm{tx}}}\boldsymbol{u}\boldsymbol{u}^H + \frac{\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^H}{N_{\mathrm{tx}}(N_{\mathrm{tx}}-1)}. \tag{63}$$

Multiplying this matrix on both sides by the true channel and using $|\boldsymbol{u}^H\boldsymbol{h}_1|^2 = Hr$ gives

$$\mathbb{E}[|\widehat{\boldsymbol{w}}^H\boldsymbol{h}_1|^2] = H\left[\frac{N_{\mathrm{tx}}-1}{N_{\mathrm{tx}}}r + \frac{1-r}{N_{\mathrm{tx}}(N_{\mathrm{tx}}-1)}\right] + o(H).$$

Finally, multiply by $\mathcal{E}_{\mathrm{D}}/N_0$ and substitute $H = N_{\mathrm{tx}}\sigma^2\gamma_1^{\mathrm{U}}$ and $(\mathcal{E}_{\mathrm{D}}/N_0)\sigma^2 = k$ to obtain (34).

B. High-SNR Limit

We prove (37) for fixed $0 \leq r < 1$. The true-channel projector defined in (56) depends only on the fixed victim-channel direction. Since $\|\boldsymbol{u}\| = 1$ and $|\boldsymbol{u}^H\boldsymbol{h}_1|^2 = Hr$,

$$\|\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{u}\|^2 = \boldsymbol{u}^H\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{u} = 1 - \frac{|\boldsymbol{u}^H\boldsymbol{h}_1|^2}{H} = 1 - r.$$

As $G_1 \to \infty$ along the fixed channel direction with σ^2 fixed, the estimated direction tends to the true direction. Equation (19) then gives, for almost every error realization,

$$\widehat{\boldsymbol{w}} \longrightarrow \boldsymbol{w}_0 = \frac{\boldsymbol{P}_{\hat{h}_1}^{\perp}\boldsymbol{u}}{\sqrt{1-r}}. \tag{64}$$

The limiting beam $\boldsymbol{w}_0$ is a deterministic unit vector. The exact error identity (22) expresses the leakage as $-\widehat{\boldsymbol{w}}^H\boldsymbol{v}_1$; its squared magnitude is bounded by the integrable variable $\|\boldsymbol{v}_1\|^2$. Dominated convergence therefore allows the limit to pass through the expectation:

$$\lim_{\gamma_1^{\mathrm{U}}\to\infty}\mathbb{E}|\widehat{\boldsymbol{w}}^H\boldsymbol{h}_1|^2 = \mathbb{E}|\boldsymbol{w}_0^H\boldsymbol{v}_1|^2 = \sigma^2. \tag{65}$$

The last equality follows from $\mathbb{E}[\boldsymbol{v}_1\boldsymbol{v}_1^H] = \sigma^2\boldsymbol{I}$ and $\|\boldsymbol{w}_0\| = 1$. Multiplying by $\mathcal{E}_{\mathrm{D}}/N_0$ and using $\sigma^2 = N_0/\mathcal{E}_{\mathrm{U}}$ proves (37).

REFERENCES

[1] S. Kang, M. Mezzavilla, S. Rangan, A. Madanayake, S. B. Venkatakrishnan, G. Hellbourg, M. Ghosh, H. Rahmani, and A. Dhananjay, "Cellular wireless networks in the upper mid-band," *IEEE Open Journal of the Communications Society*, 2024.

[2] D. Shakya, M. Ying, T. S. Rappaport, H. Poddar, P. Ma, Y. Wang, and I. Al-Wazani, "Comprehensive FR1(C) and FR3 Lower and Upper Mid-Band Propagation and Material Penetration Loss Measurements and Channel Models in Indoor Environment for 5G and 6G," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 5192–5218, 2024.

[3] National Telecommunications and Information Administration, "World radiocommunication conference 2023 outcomes and implications for u.s. spectrum policy," 2024, report on WRC-23 outcomes regarding FR3 upper mid-band identification.

[4] F. C. Commission *et al.*, "Single network future: Supplemental coverage from space," 2024.

[5] N. R. Olson, T. S. Rappaport, and J. G. Andrews, "A tractable framework for spectrum coexistence between satellite receivers and terrestrial networks," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 6758–6779, 2024.

[6] S. Kang, G. Geraci, M. Mezzavilla, and S. Rangan, "Terrestrial-satellite spectrum sharing in the upper mid-band with interference nulling," in *ICC 2024 - IEEE International Conference on Communications*, 2024, pp. 5057–5062.

[7] S. K. Sharma, S. Chatzinotas, J. Grotz, and B. Ottersten, "3D beamforming for spectral coexistence of satellite and terrestrial networks," in *2015 IEEE 82nd Vehicular Technology Conference (VTC2015-Fall)*. IEEE, 2015, pp. 1–5.

[8] S. K. Sharma, S. Chatzinotas, and B. Ottersten, "Transmit beamforming for spectral coexistence of satellite and terrestrial networks," in *8th International Conference on Cognitive Radio Oriented Wireless Networks*. IEEE, 2013, pp. 275–281.

[9] Z. Lin, M. Lin, J.-B. Wang, Y. Huang, and W.-P. Zhu, "Robust secure beamforming for 5G cellular networks coexisting with satellite networks," *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 4, pp. 932–945, 2018.

[10] J. Lim and J. Kang, "Interference analysis for coexistence of satellite and terrestrial networks in shared spectrum," *IEEE Communications Letters*, vol. 27, no. 2, pp. 513–517, 2023.

[11] H. Zhang, C. Jiang, J. Wang, L. Wang, Y. Ren, and L. Hanzo, "Multicast beamforming optimization in cloud-based heterogeneous terrestrial and satellite networks," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 2, pp. 1766–1776, 2019.

[12] E. Kim, I. P. Roberts, and J. G. Andrews, "Feasibility analysis of in-band coexistence in dense leo satellite communication systems," *IEEE Transactions on Wireless Communications*, 2024.

[13] E. Kim, I. P. Roberts, T. Lee, and J. G. Andrews, "Satellite selection for in-band coexistence of dense leo networks," *arXiv preprint arXiv:2503.15262*, 2025.

[14] H.-W. Lee, A. Medles, C.-C. Chen, and H.-Y. Wei, "Feasibility and opportunities of terrestrial network and non-terrestrial network spectrum sharing," *IEEE Wireless Communications*, vol. 30, no. 6, pp. 36–42, 2023.

[15] H.-W. Lee, C.-C. Chen, C.-I. S. Liao, A. Medles, D. Lin, I.-K. Fu, and H.-Y. Wei, "Interference mitigation for reverse spectrum sharing in B5G/6G satellite-terrestrial networks," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 3, pp. 4247–4263, 2023.

[16] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022, https://nvlabs.github.io/sionna/.

[17] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz." Tech. Rep. TR 38.821, 2025.